



Transforming Data into Gold: A Live Demonstration of Microsoft Fabric Lakehouse

Data Saturday, Oslo, 30/8-2025

twoday



Thank you to our sponsors



DATA SATURDAY OSLO



Measure Killer



beta systems

Agenda

Fabric is Fantastic -
but also
overwhelming



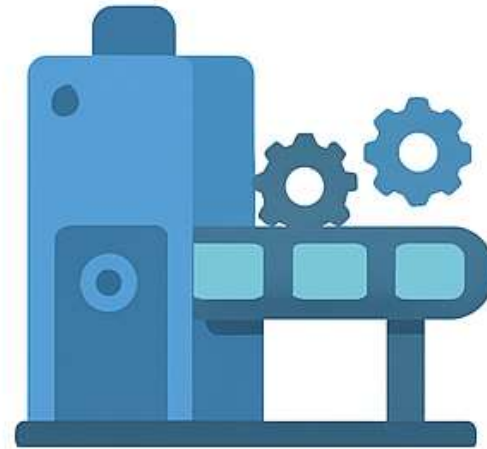
Lakehouse
starting-point
architecture



Kickstart
the
Lakehouse
Setup



Start the
Lakehouse
with an
accelerator



Extend and
customize the
accelerator



Just Blindbæk

- Principal Architect at twoday
 - Pre-sales and marketing
 - Internal practice development
 - Academy: External training
- Microsoft Data Platform MVP
- Found and organizer of
 - Danish Microsoft BI Community (MsBIP.dk)
 - Power BI UG Denmark (PowerBI.dk)
 - Power BI Next Step and Data Platform Next Step

 [linkedin.com/in/blindbaek/](https://www.linkedin.com/in/blindbaek/)

twoday

(Christian) Henrik Reich

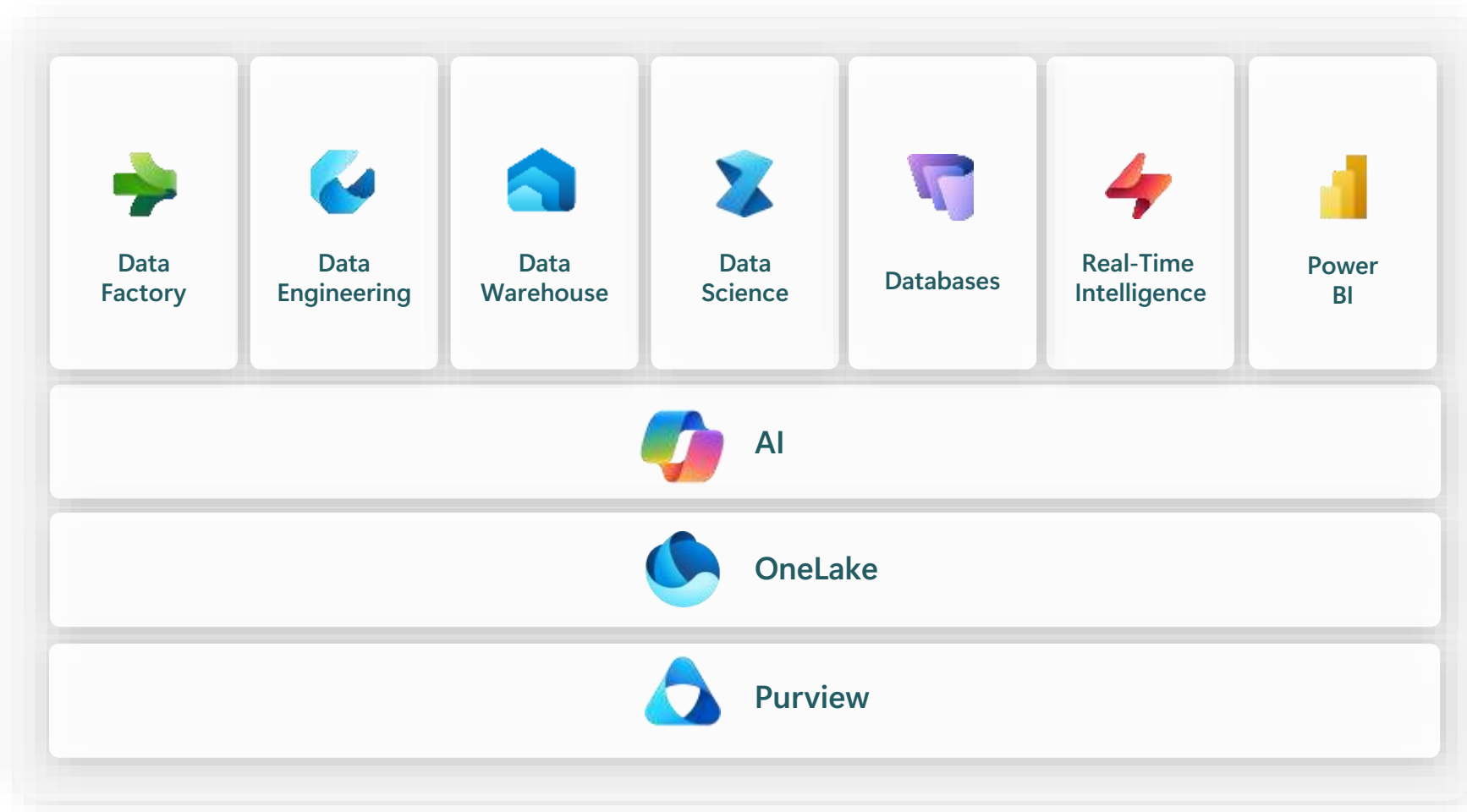


- Principal Architect @ twoday Data & AI (Technology And Architecture)
- Redi-School Teacher. Teaching Microsoft AI Technologies to immigrant women. Starting 5th semester.
- Ex-Pluralsight assessment tests author
- Øredev program committee member since 2019 (Data track) (Malmø)
- Agile Global Summit 2025 program committee member (Tallinn)
- Pass Summit program committee member 2023+2025 (Seattle)
- Holding 23 active and 3 historical microsoft certificates
- Been a developer in many niches

 [linkedin.com/in/henrik-reich](https://www.linkedin.com/in/henrik-reich)
medium.com/@christianhenrikreich

twoday

Fabric: End-to-end unified analytical platform



Software as a Service

Seven workloads

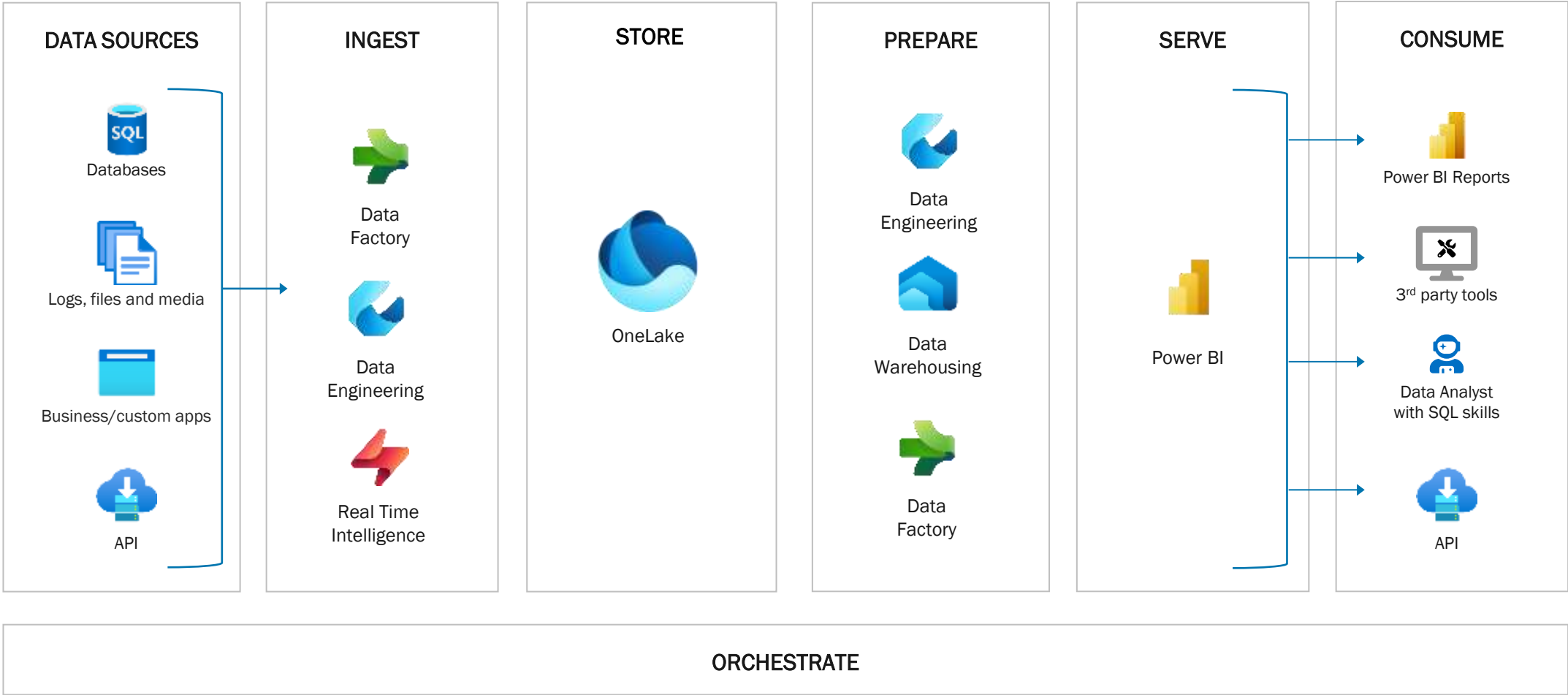
- Low-code/no-code
- Pro developer / engineers

- 30+ items

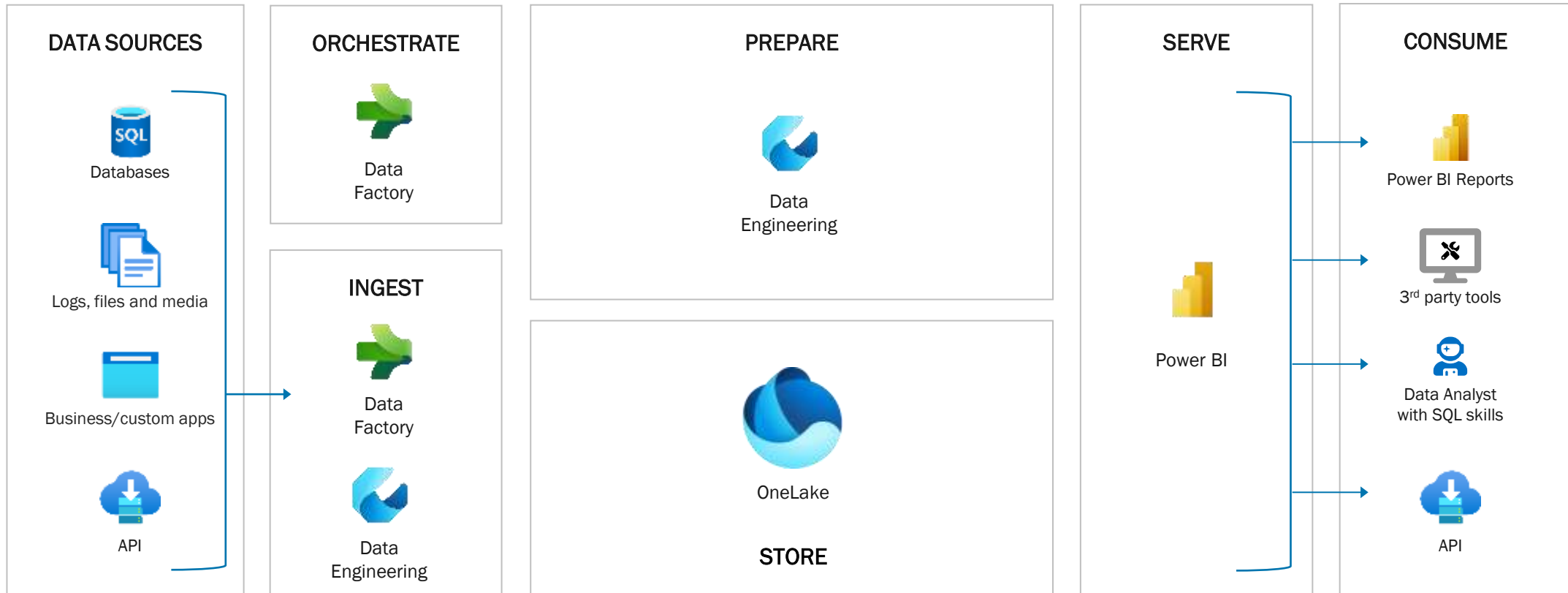
OneLake & Delta format

- Separation of storage and compute

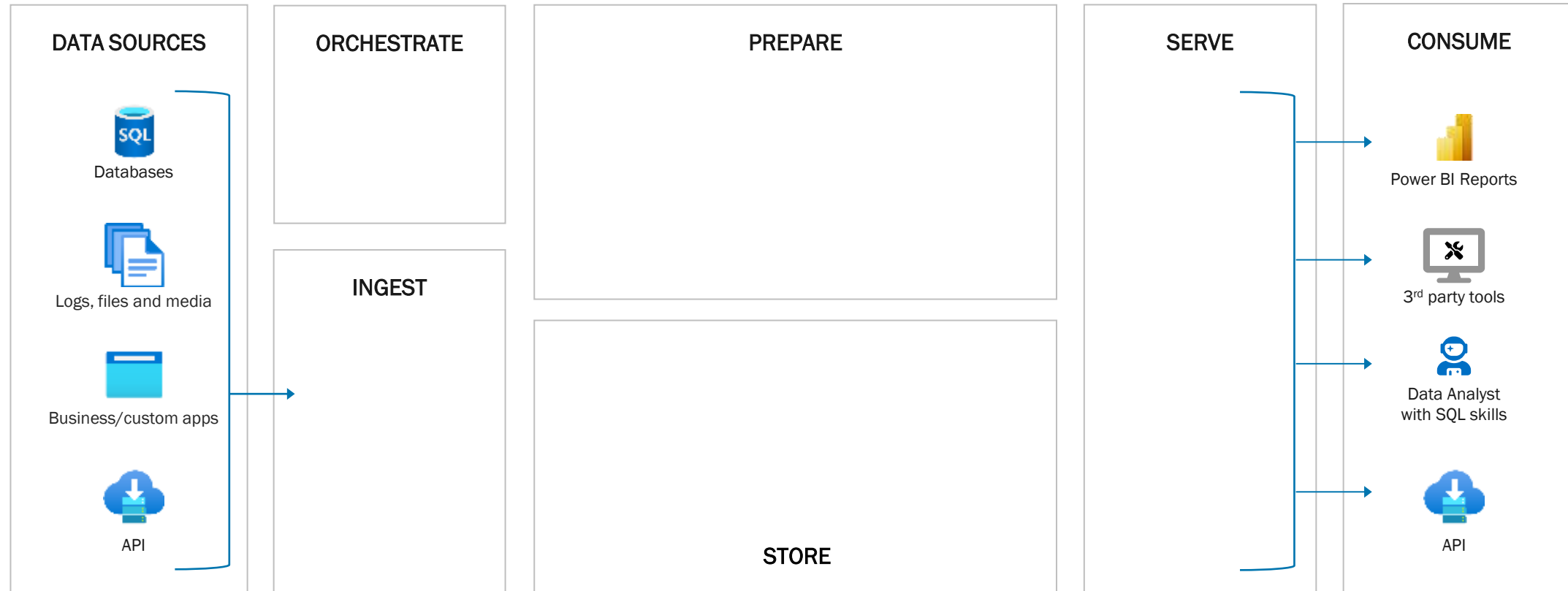
Architecture view of Fabric workloads



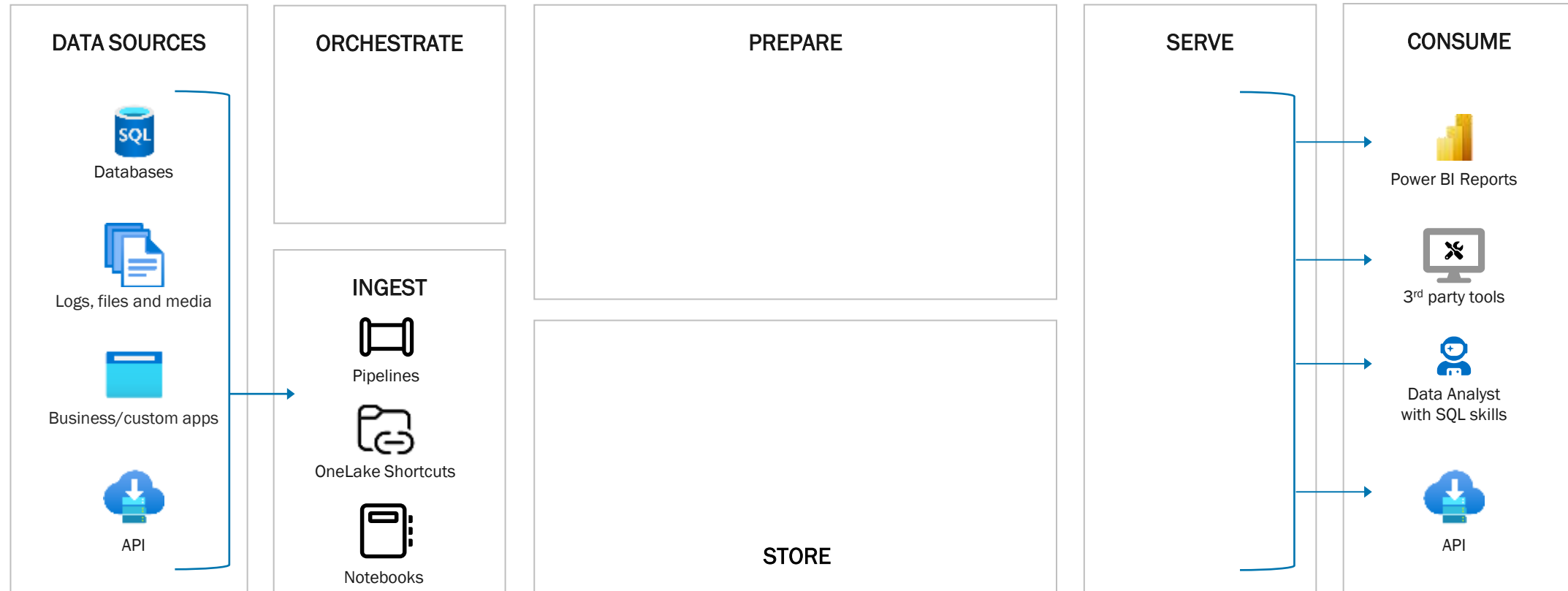
Lakehouse view of Fabric workloads



Dataflow in a Data Lakehouse in Fabric



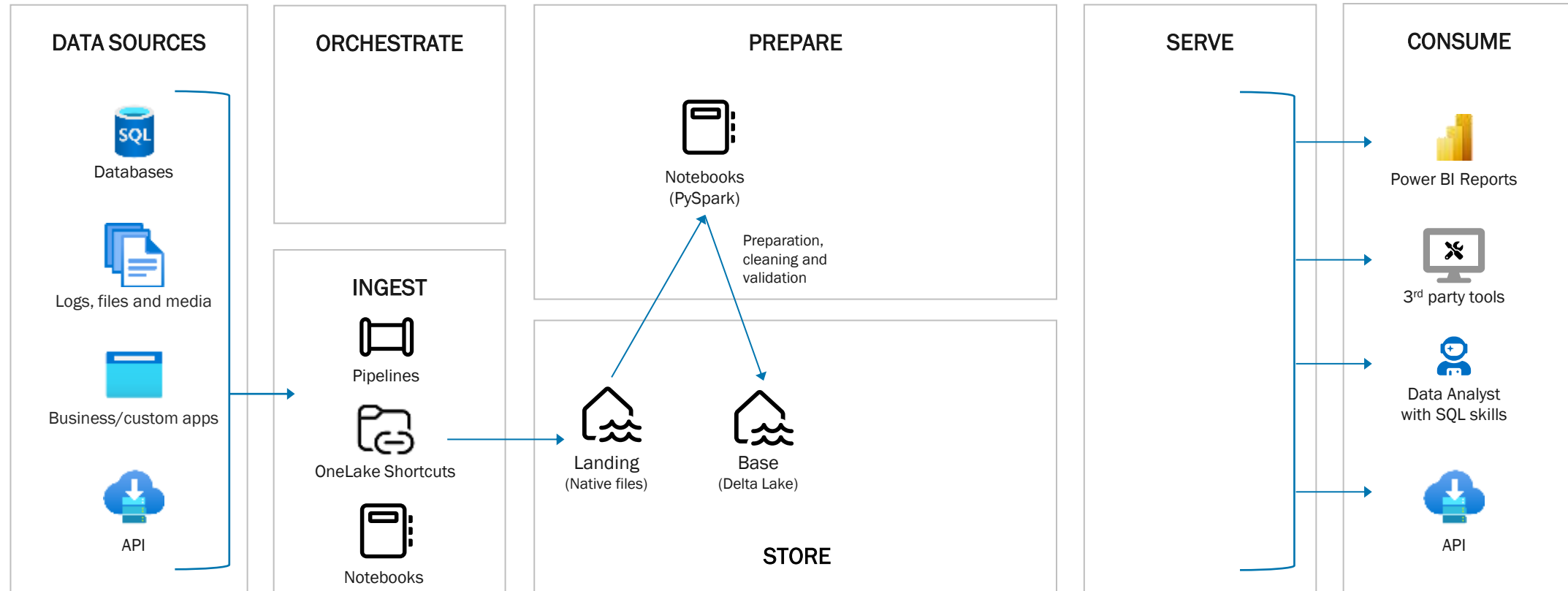
Dataflow in a Data Lakehouse in Fabric



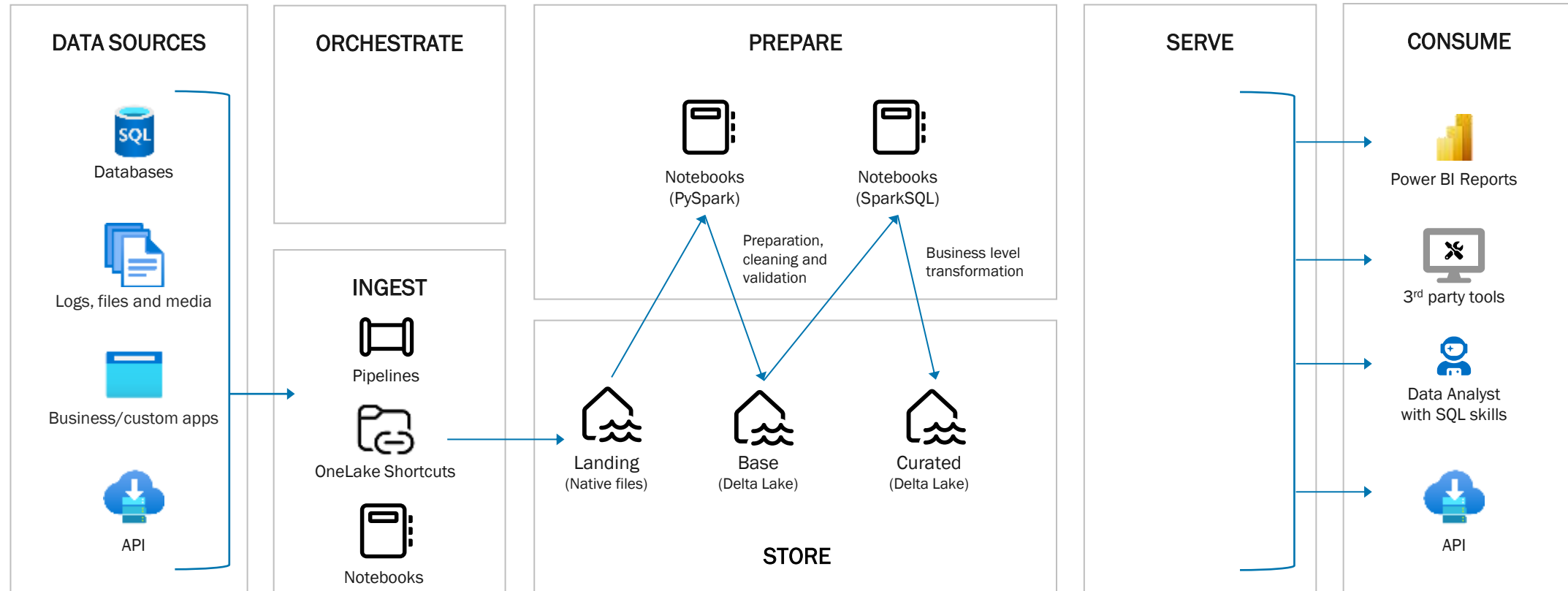
Dataflow in a Data Lakehouse in Fabric



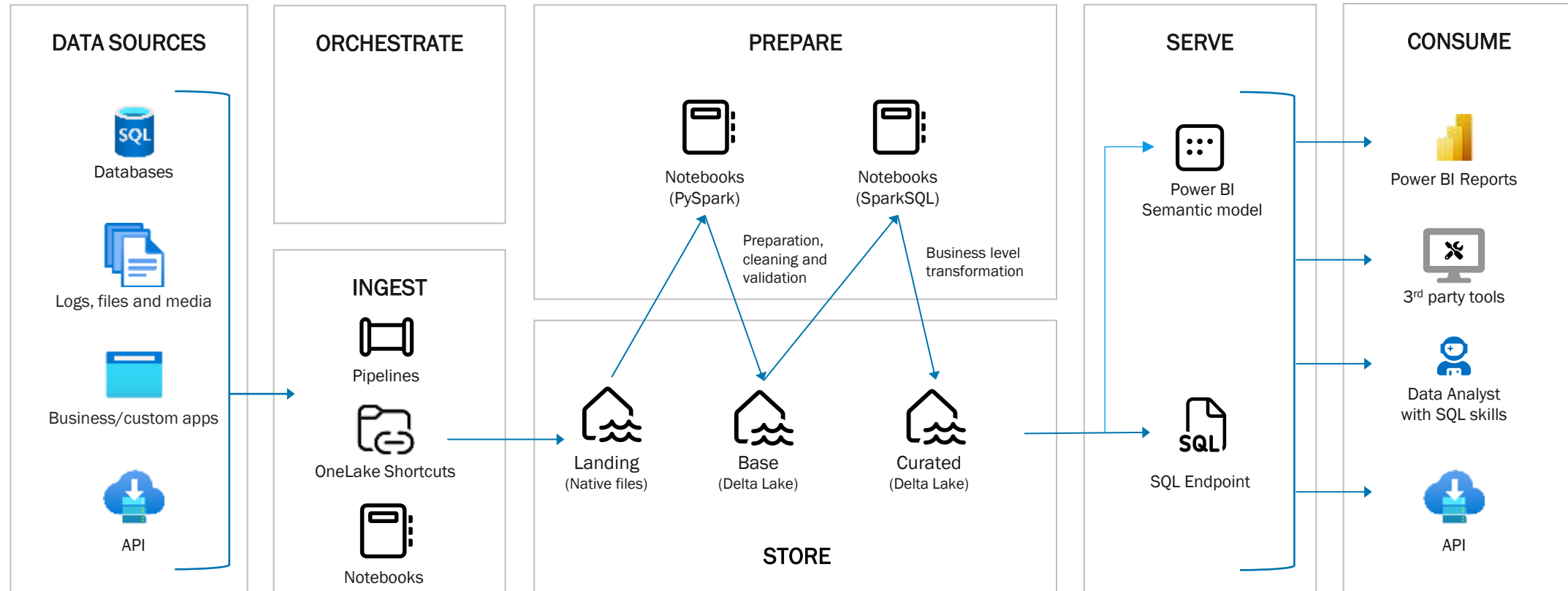
Dataflow in a Data Lakehouse in Fabric



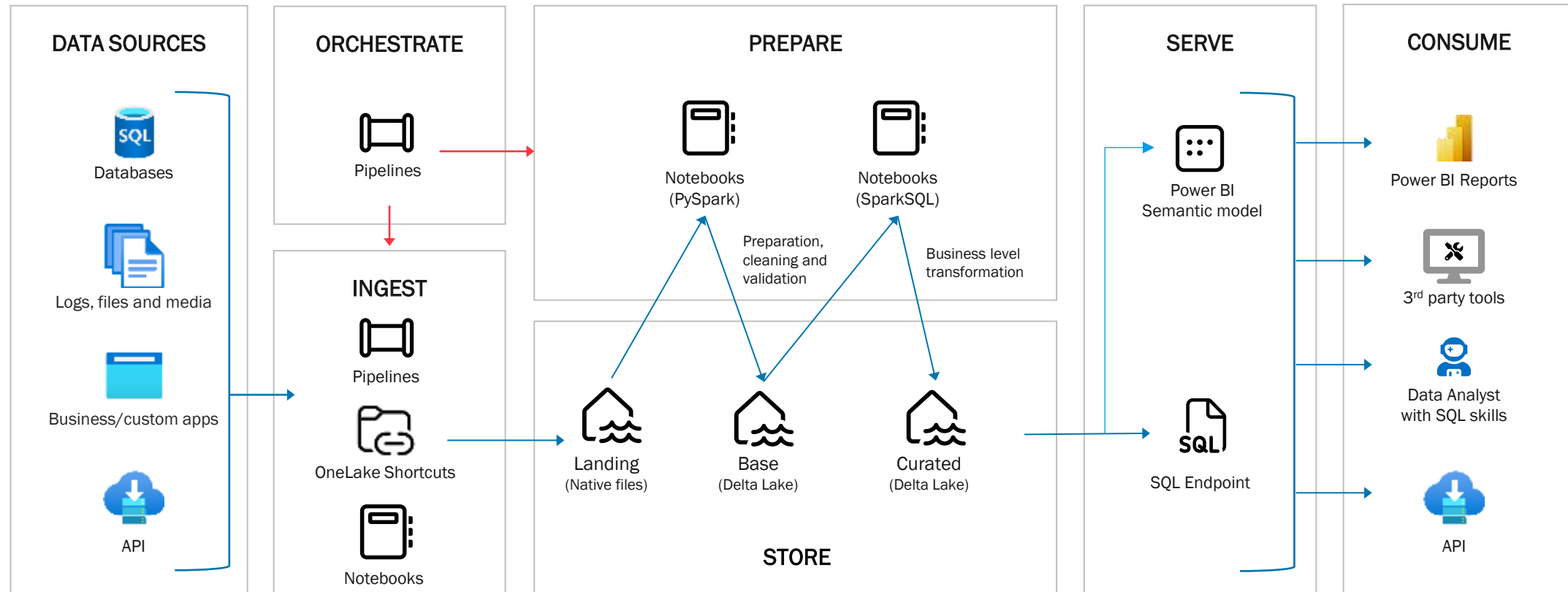
Dataflow in a Data Lakehouse in Fabric



Dataflow in a Data Lakehouse in Fabric



Dataflow in a Data Lakehouse in Fabric

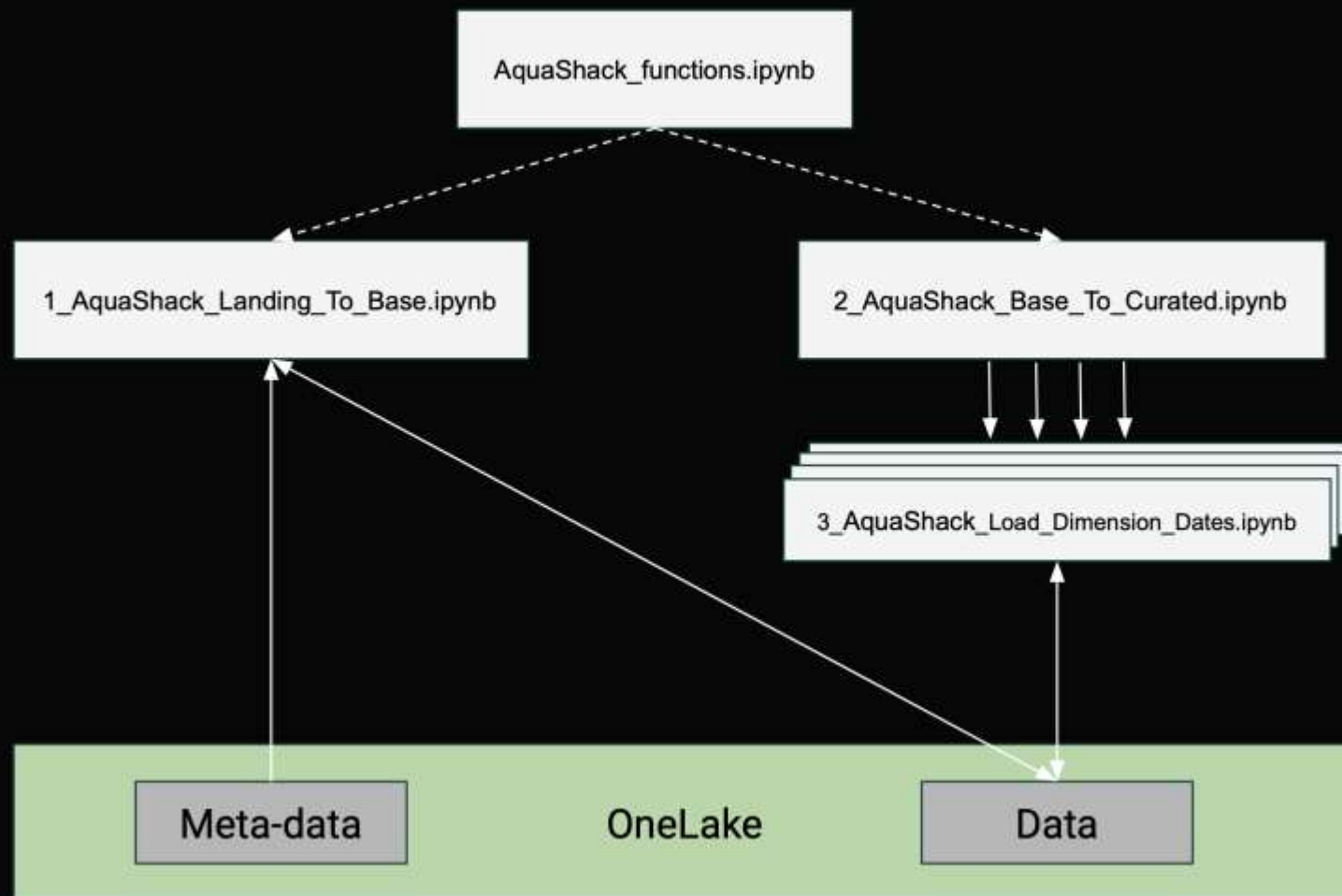


AquaShack: Data Lakehouse accelerator

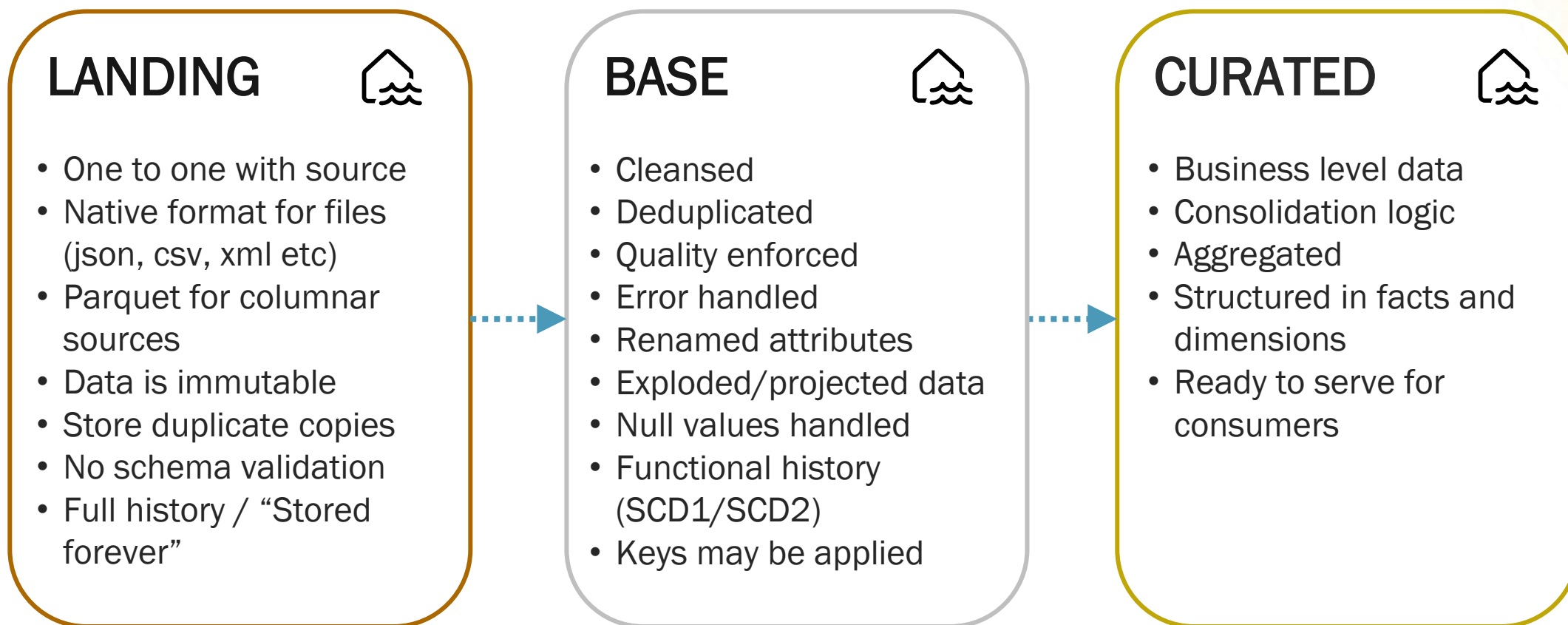


- Pico-example of a meta-data driven lakehouse for Microsoft. Little brother to AquaVilla
- Data Lakehouse with three layers, where data is moved between.
 - Landing: Holds the data in original format if possible. Relation sources are stored in parquet.
 - Base: Aligned clean data, all stored in Delta tables.
 - Curated: Data for serving, business logic is applied, star schemas are defined etc.
- The notebooks builds upon functions defined in **AquaShack_functions**.
- We are reading metadata from JSON, it could have been from a database or YAML file as well. The important part is it ends in a common structure.

AquaShack



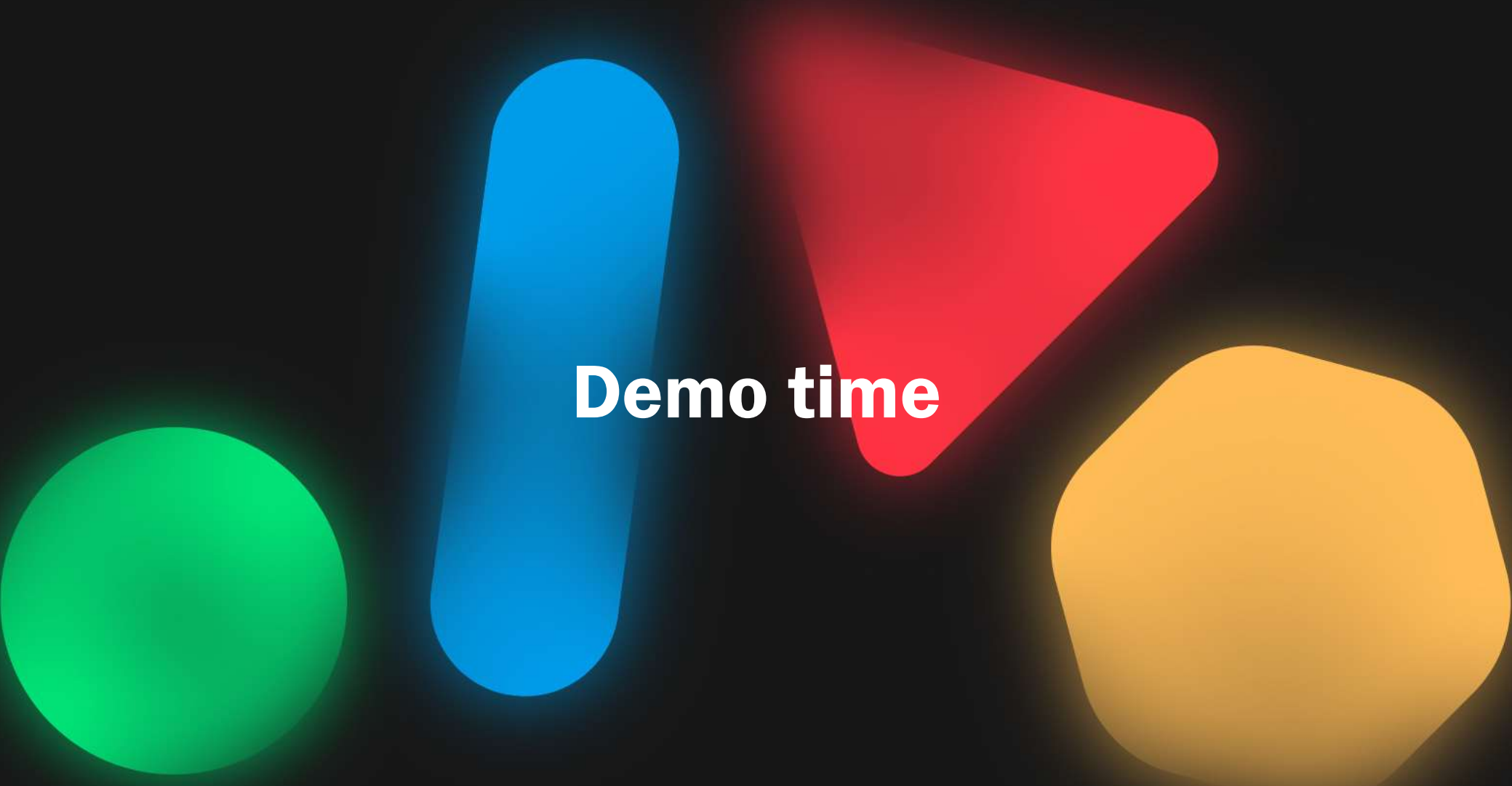
Layers in the Data Lakehouse



Metadata driven from Landing to Base

- The **AquaShack_Functions** notebook contains general functions used for operations like reading DataFrames, writing DataFrames, and utilities for manipulating DataFrames. To use the functions, include the notebook in your Spark session with the %run command. Once included, you can reference any function within the notebook, such as read_meta_from_json, write_to_delta_overwrite, and others.
- Data movement between the Landing and Base layers is managed by the LandingToBase notebook. This notebook reads metadata from a designated json-file and processes the source path (folder) specified within it.
- Metadata is maintained in the meta_data.json file. This metadata defines the rules for reading data from the Landing layer and writing it to the Base layer.

```
[
  {
    "source": "Files/External/Sample data/Sales/Customers/Customers.csv",
    "format": "csv",
    "destination": "sales_Customers",
    "projected_columns": ["CustomerID", "FirstName", "LastName"]
  },
  {
    "source": "Files/External/Sample data/Sales/Products/Products.parquet",
    "format": "parquet",
    "destination": "sales_Products",
    "projected_columns": []
  },
  {
    "source": "Files/External/Sample data/Sales/Transactions/",
    "format": "csv",
    "destination": "sales_Transactions",
    "projected_columns": []
  }
]
```



Demo time

twoday

Base to Curated

- The Curated layer stores data in a dimensional model with entities like dimensions, facts, and bridge tables.
- Entity-specific Spark notebooks handle data creation, named as:
 - Load_Dimension_[dimension_name]
 - Load_Fact_[fact_name]
- Each notebook follows a standard structure:
 - Parameters & Settings.
 - Business Logic: Written in SparkSQL.
 - Generic Load Execution: Functions like load_dimension or load_fact.
- All notebooks are tied to the Base Lakehouse item for centralized management.

Defining settings

Some settings run through the notebook and easy to set once before the job starts.

```
1 destination_lakehouse: str = "Curated"
2
3 dimension_name: str = "Product"
```

Defining business requirements

This could also be pure PySpark, it depends on the skill set of the BI Developer.

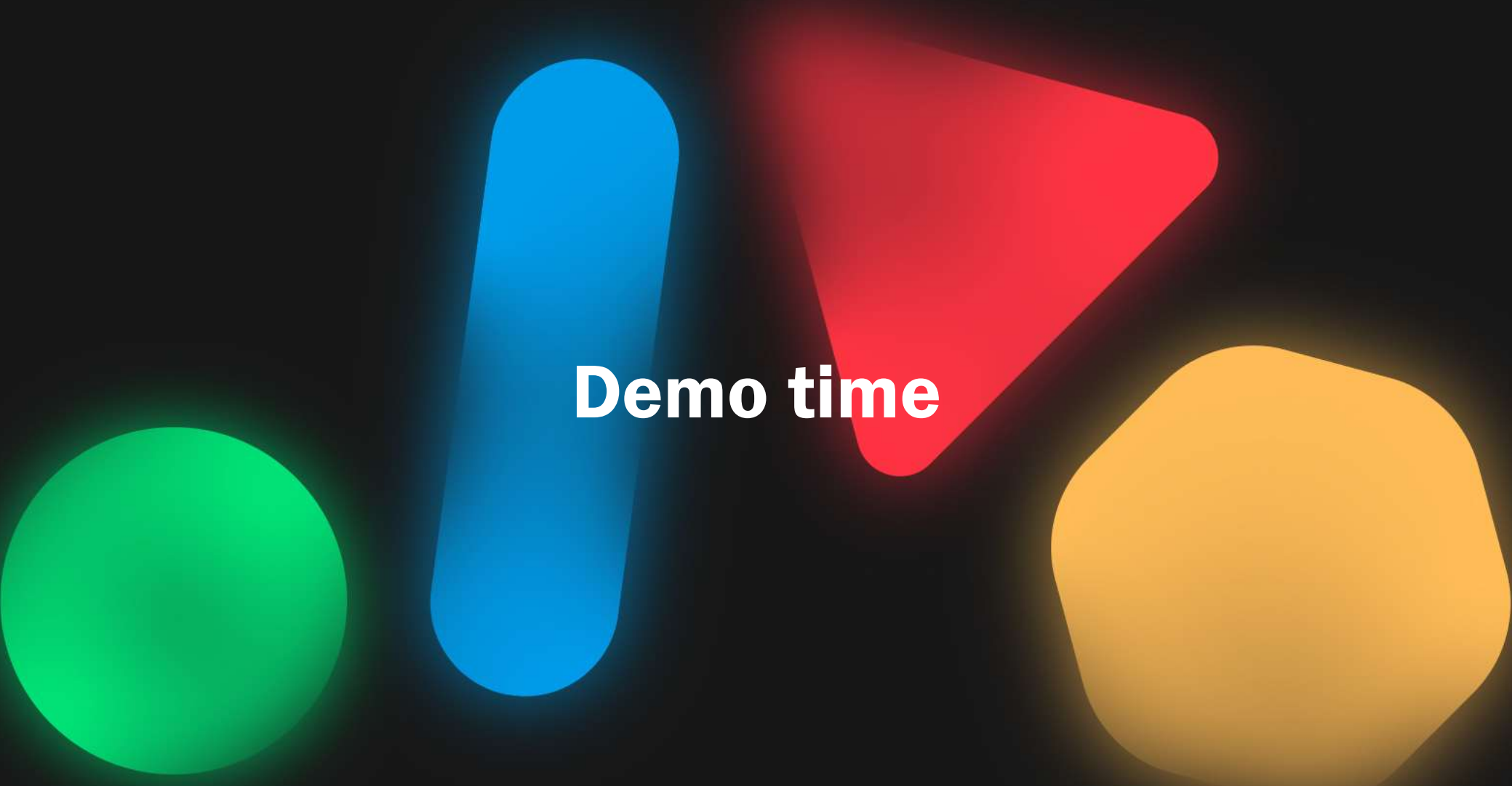
```
1 -- Step 1
2 CREATE OR REPLACE TEMPORARY VIEW dim_df AS
3
4 SELECT
5     -- Keys
6     Id as Dim_ProductId,
7
8     -- Subcategory
9     ProductId as ProductSubId,
10
11     -- Attributes
12     Manufacturer,
13     ProductName,
14     Price
15
16 FROM
17     Base.catalog_products
```

Writing the dimension

Fetch the DataFrame and pass it to our function for writing dimensions.

```
1 dim_df = spark.table('dim_df')
2
3 load_dimension(df = dim_df, database = destination_lakehouse, table = dimension_name)
```

Overwriting Curated-dim_catalog_products



Demo time

twoday

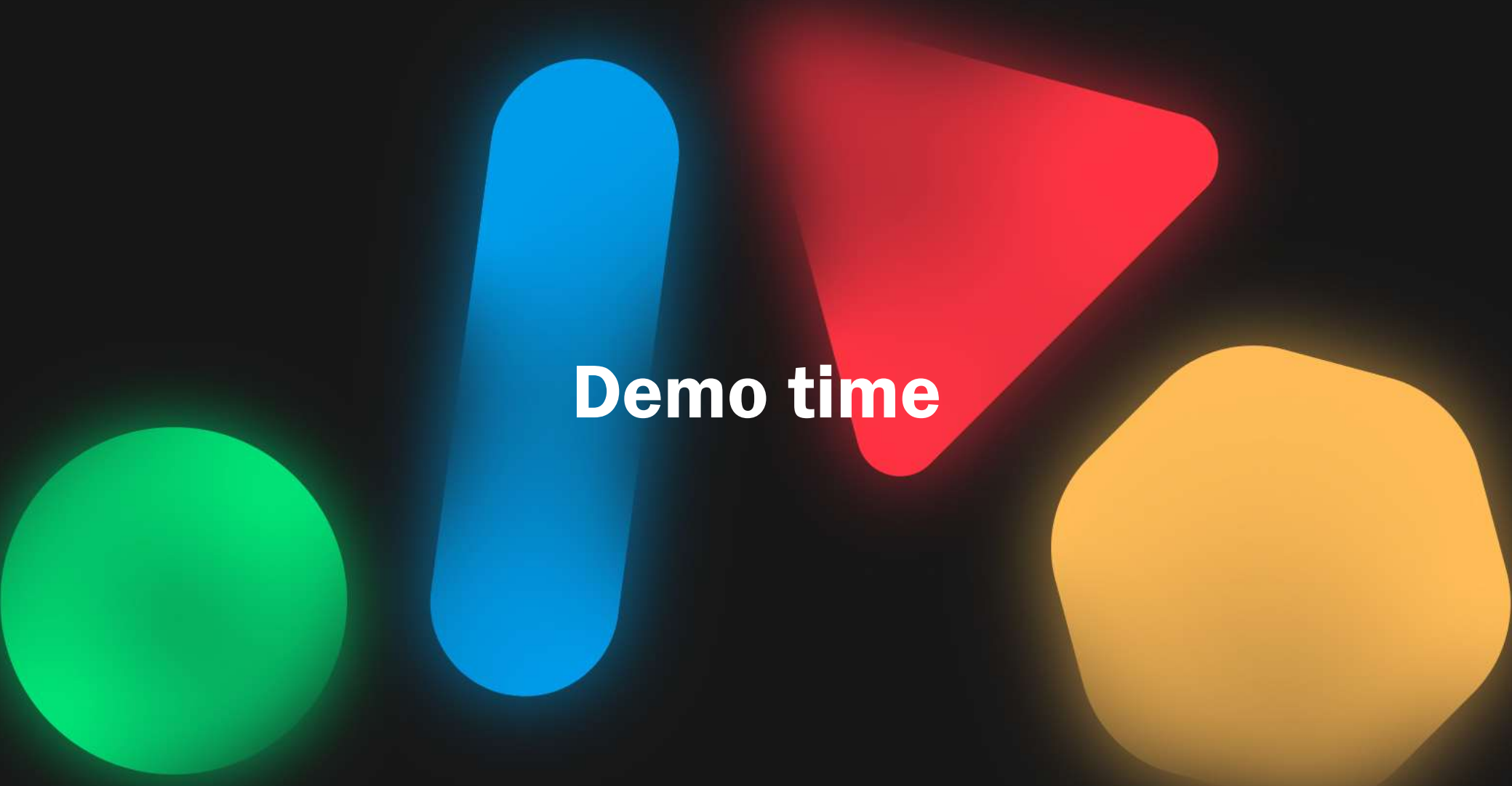
Quick start guide to AquaShack



- **Go to GitHub:** <https://github.com/ChristianHenrikReich/AquaShack>
- Download or copy the Setup.ipynb notebook that holds all to get started and set up the AquaShack lakehouse example.
- The only required action is to import and run the notebook within the target Fabric Workspace.
- The setup notebook will create all you need:
 - The three Lakehouses
 - All the Notebooks – attached to default lakehouses
 - Sample metadata file
 - Sample data in Landing



twoday



Demo time

twoday

Extend and Customize the Accelerator

- AquaShack is a pico example - a small but powerful starting point.
- Teach and get started with
 - The medallion architecture
 - Lakehouse concepts
 - Hands-on Spark development in Fabric
- Get you going fast and help you build good habits from day one



twoday

Extend with more features

- **Automated setup:** Full setup of infrastructure, workspaces and items
- **Metadata:** Integration, processing and orchestration with metadata
- **Ingestion:** Batch ingest from multiple sources using delta and full.
- **Common transformations:** Deduplication, flattening of complex data structures, data conversion, translations and more.
- **Update strategies:** Append, overwrite, upsert type 1 & 2 etc.
- **Data warehousing:** Functions specifically for dimensional modelling
- **Samples:** Sample implementations for all functionality and layers
- **Orchestration:** Through Pipelines and Notebooks
- **Deployment:** Full enterprise CI/CD using Azure DevOps Pipelines





Extend with more modules

Infrastructure

Publisher app and scripts

- Azure infrastructure
- Fabric env. setup



Solution

Metadata-driven ingest, transform and orchestration

- Data Pipelines
- Notebooks
- Onelake as storage
- Metadata-driven development



CI/CD

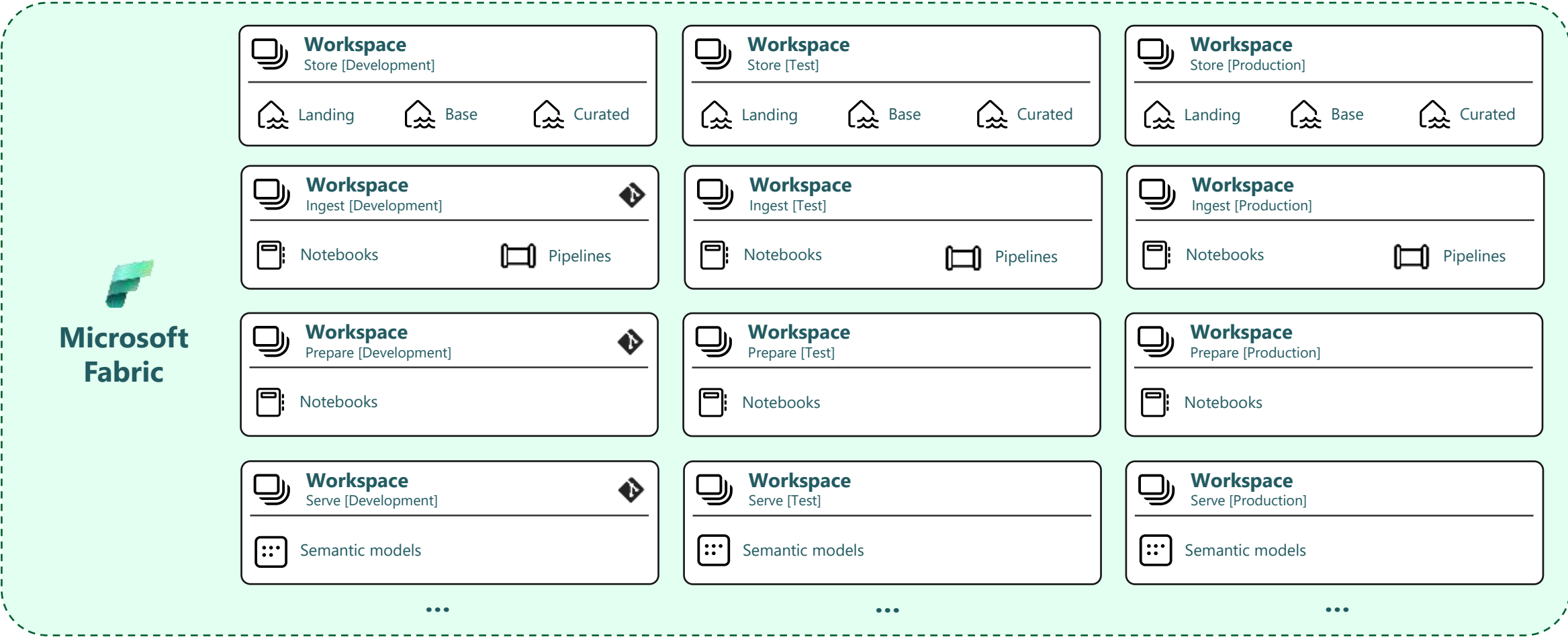
Pipelines and scripts

- Build and release
- Handling features



Documentation, learning and guidelines

Platform Infrastructure with Fabric workspaces



Setup the Fabric Lakehouse structure in 1 minute!

Initialization script & recipe

Using Service Principal (SPN)

- Create workspaces
- Assign workspace permissions
- Assign capacities to workspaces
- Create Fabric items (lakehouses)
- Create managed private endpoints

Using User principal (UPN)*

- Connects workspaces to git
- Initializes Git connection
- Updates workspaces from Git

Fabric REST APIs

Core APIs

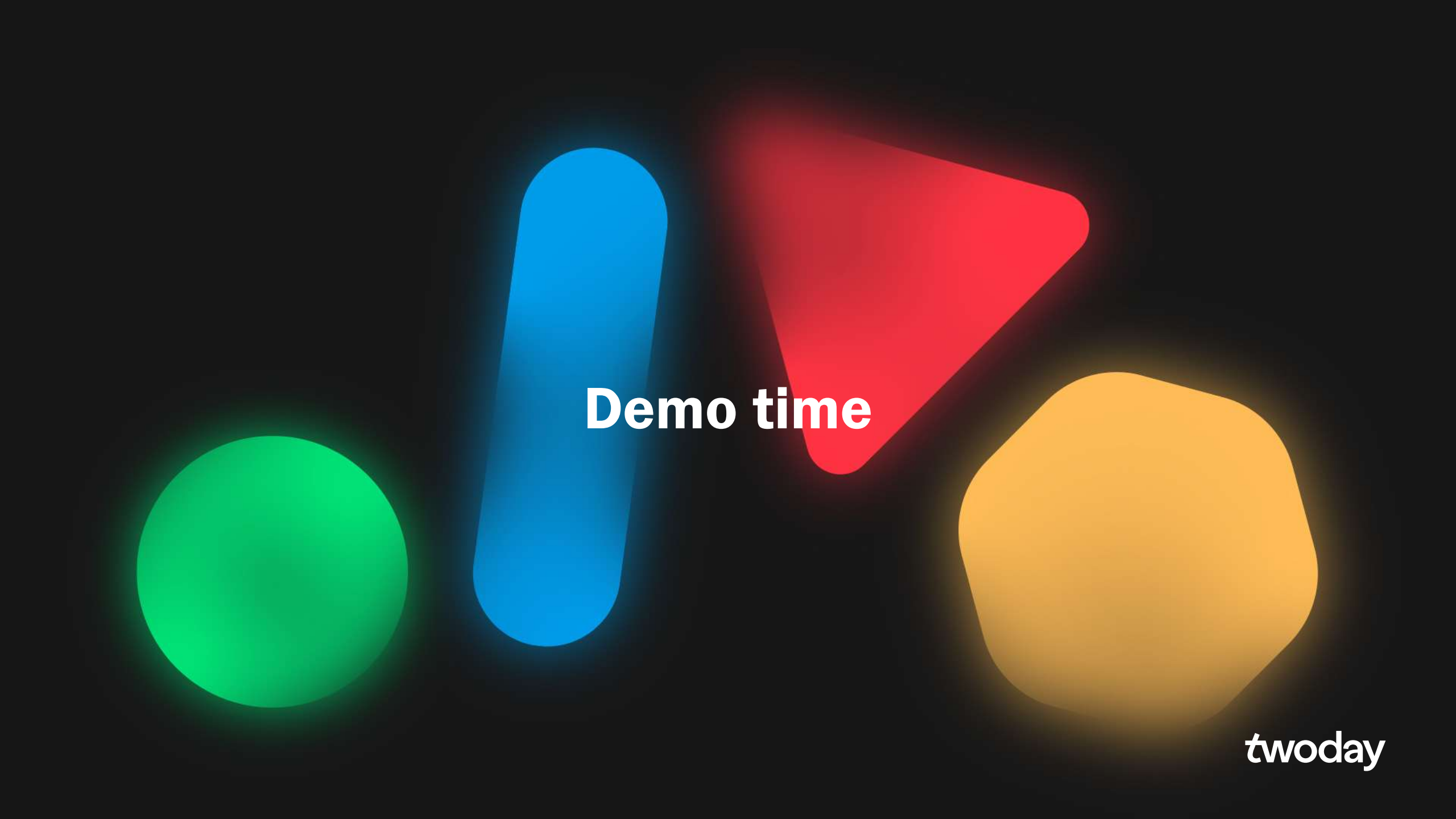
- Workspaces
- Items
- Long Running Operations
- Managed private endpoints
- Git*



Visit Peer's blog post "Automating Fabric: Kickstart your Fabric Setup with Python and Fabric REST APIs" on <https://peerinsights.hashnode.dev>



twoday



Demo time

twoday

Key takeaways

- **Automation with PySpark**
 - Clean, validate, and deduplicate data with reusable functions
- **Business Logic in SQLSpark**
 - SQL makes your models portable and easy to maintain.
- **Open-Source Productivity**
 - Build with our framework to save time and raise quality.
- **Code You Can Reuse**
 - Try AquaShack yourself via GitHub after the session.



twoday



Time for questions!

twoday

Evaluate this session



<https://eval.datasaturday.no/190773860634797>

twoday



Thank you to our sponsors



DATA SATURDAY OSLO



Measure Killer





twoday